

## **B.Sc. (Hons.) in Environmental Science**

Environmental Science is an interdisciplinary field that studies the interactions between living and non-living components of the environment. The B.Sc. (Hons.) in Environmental Science program is designed to provide students with a comprehensive understanding of various environmental issues and equip them with skills and knowledge to tackle these issues. The program is typically spread over three years and covers a range of topics related to environment and sustainability.

The following are the key areas of study in the B.Sc. (Hons.) in Environmental Science program:

### **1. Environmental Chemistry**

- Study of chemical composition of the environment and its impact on living organisms.
- Examination of the sources, transport, fate and effects of pollutants in air, water and soil.
- Understanding of chemical reactions that occur in the environment.

### **2. Environmental Biology**

- Study of the interactions between living organisms and their environment.
- Examination of the impact of environmental factors on the growth and survival of organisms.
- Understanding of the ecological principles that govern the functioning of ecosystems.

### **3. Environmental Physics**

- Study of physical aspects of the environment such as climate, weather and energy.
- Examination of the impact of human activities on the physical environment.
- Understanding of the principles of energy conversion and the role of renewable energy sources.

### **4. Environmental Policy and Management**

- Study of the policies and regulations related to environmental management and sustainability.
- Examination of the role of government, NGOs and other stakeholders in environmental decision-making.
- Understanding of the tools and techniques for environmental monitoring and assessment.

## **5. Environmental Impact Assessment**

- Study of the methods and tools used to assess the potential impact of human activities on the environment.
- Examination of the legal and regulatory frameworks for environmental impact assessment.
- Understanding of the role of stakeholders in the environmental impact assessment process.

## **6. Environmental Ethics and Philosophy**

- Study of the ethical and philosophical dimensions of environmental issues.
- Examination of the different ethical frameworks for environmental decision-making.
- Understanding of the role of culture and values in shaping environmental attitudes and behaviors.

In addition to the above areas of study, the B.Sc. (Hons.) in Environmental Science program may also cover topics such as climate change, biodiversity conservation, sustainable development, and environmental education.

Upon completion of the program, graduates can pursue careers in a range of fields such as environmental consulting, environmental policy, sustainable development, conservation, and education. They can also pursue further studies in related fields such as environmental law, environmental engineering, and ecology.

## **Unit 1 - Environment and Ecosystem**

### **Environment**

Environment is the sum total of all living and non-living things that surround us. It includes all the natural resources, such as air, water, soil, and living organisms, as well as human-made structures and objects. The environment can be broadly classified into two types:

1. Natural Environment: The natural environment consists of all the living and non-living things that occur naturally in the world, without any human intervention. It includes things like forests, oceans, rivers, mountains, and wildlife.
2. Human-made Environment: The human-made environment is made up of all the things that humans have created, such as buildings, roads, bridges, and cities.

The environment has several components that work together to form a complex web of life. These components can be broadly classified into four segments:

1. Atmosphere: This segment includes the air we breathe, the weather patterns, and the climate.

2. Hydrosphere: This segment includes all the water bodies on the planet, such as oceans, rivers, lakes, and groundwater.
3. Lithosphere: This segment includes the solid, outermost layer of the Earth, which includes the soil, rocks, minerals, and landforms.
4. Biosphere: This segment includes all the living organisms on the planet, including plants, animals, and microorganisms.

The environment plays a crucial role in sustaining life on the planet. It provides us with all the resources we need to survive, such as food, water, and air. Therefore, it is important to understand the scope and importance of the environment and the need for public awareness to protect it.

### **Ecosystem**

An ecosystem is a community of living and non-living things that interact with each other in a particular environment. It includes all the living organisms, such as plants, animals, and microorganisms, as well as non-living things, such as air, water, and soil.

There are two types of ecosystems:

1. Natural Ecosystem: A natural ecosystem is one that exists without any human intervention, such as a forest or a coral reef.
2. Artificial Ecosystem: An artificial ecosystem is one that is created by humans, such as a park or a garden.

The structure of an ecosystem is based on the food chain, which shows the flow of energy from one organism to another. The food chain begins with the primary producers, such as plants, which convert sunlight into energy through photosynthesis. The primary producers are then eaten by herbivores, which are then eaten by carnivores.

The food chain is interconnected and forms a complex web of life, known as the food web. The food web includes all the organisms in an ecosystem and shows how they are connected to each other.

The ecological pyramid is a graphical representation of the energy flow in an ecosystem. It is divided into three levels: the producers, the consumers, and the decomposers. The producers are at the bottom of the pyramid, while the consumers and decomposers are at the top.

It is important to maintain a balance in the ecosystem to ensure the survival of all the organisms within it. Human activities, such as food, shelter, housing, agriculture, industry, mining, transportation, economic and social security, can have a significant impact on the environment. Therefore, it is important to conduct an environmental impact assessment before undertaking any such activities.

Sustainable development is a way of using resources in a manner that meets the needs of the present without compromising the ability of future generations to meet their own needs. It is important to adopt sustainable development practices to ensure the long-term survival of the environment and its components.

## **Unit 2 - Natural Resources: Introduction, Classification**

In this unit, we will be discussing a variety of natural resources that are essential to our survival and the sustainability of our planet. We will be covering the following topics:

### **Water Resources**

- Availability of water resources
- Sources of water
- Quality aspects of water
- Water-borne diseases
- Water-induced diseases
- Fluoride and arsenic problems in drinking water

### **Mineral Resources**

- Material cycles
- Carbon cycle
- Nitrogen cycle
- Sulfur cycle

### **Energy Resources**

- Conventional sources of energy
- Non-conventional sources of energy

### **Forest Resources**

- Availability of forest resources
- Depletion of forests
- Environmental impact of forest depletion on society

Let's dive into each of these topics in more detail.

## **Water Resources**

Water is one of the most important natural resources that we have. It is essential for our survival and for the sustainability of our planet. In this section, we will be discussing various aspects of water resources.

### **Availability of Water Resources**

- Only 3% of the world's water is freshwater and most of it is not accessible for human use
- Groundwater and surface water are the two main sources of freshwater
- Availability of freshwater varies by region and season

### **Sources of Water**

- Surface water sources include rivers, lakes, and reservoirs
- Groundwater sources include aquifers and wells

### **Quality Aspects of Water**

- Water quality can be influenced by natural and human factors
- Common contaminants include bacteria, viruses, and chemicals
- Safe drinking water standards have been established by the World Health Organization (WHO)

### **Water-borne and Water-induced Diseases**

- Water-borne diseases are caused by consuming contaminated water
- Water-induced diseases are caused by exposure to water, such as through swimming or bathing
- Common water-borne and water-induced diseases include cholera, typhoid, and malaria

### **Fluoride and Arsenic Problems in Drinking Water**

- High levels of fluoride and arsenic in drinking water can cause health problems
- Fluorosis is a condition caused by excessive fluoride consumption
- Arsenic poisoning can cause cancer and other health problems

### **Mineral Resources**

Mineral resources are an important natural resource that are used in a variety of applications. In this section, we will be discussing material cycles and the carbon, nitrogen, and sulfur cycles.

#### **Material Cycles**

- Material cycles are the processes by which minerals are recycled through the environment
- Material cycles include the rock cycle, the water cycle, and the carbon, nitrogen, and sulfur cycles

### **Carbon Cycle**

- The carbon cycle is the process by which carbon is cycled through the environment
- Carbon is released into the atmosphere through the burning of fossil fuels and other human activities
- Carbon is absorbed by plants through photosynthesis

### **Nitrogen Cycle**

- The nitrogen cycle is the process by which nitrogen is cycled through the environment
- Nitrogen is essential for plant growth
- Nitrogen can be fixed by bacteria in the soil or through lightning strikes

### **Sulfur Cycle**

- The sulfur cycle is the process by which sulfur is cycled through the environment
- Sulfur is an essential element for plant growth
- Sulfur can be found in rocks and minerals

## **Energy Resources**

Energy resources are essential for our daily lives and for the sustainability of our planet. In this section, we will be discussing conventional and non-conventional sources of energy.

### **Conventional Sources of Energy**

- Conventional sources of energy include fossil fuels such as coal, oil, and natural gas
- These sources of energy are finite and have a negative impact on the environment

### **Non-conventional Sources of Energy**

- Non-conventional sources of energy include renewable energy sources such as solar, wind, and hydropower
- These sources of energy are sustainable and have a lower impact on the environment

## **Forest Resources**

Forests are an important natural resource that provide a variety of benefits to society. In this section, we will be discussing the availability of forest resources, forest depletion, and the environmental impact of forest depletion on society.

### **Availability of Forest Resources**

- Forests provide a variety of benefits, including timber, recreation, and habitat for wildlife
- Forests cover approximately 30% of the Earth's land area

### **Depletion of Forests**

- Deforestation is the process by which forests are cleared for human use
- Deforestation has a variety of negative impacts on the environment, including soil erosion, loss of biodiversity, and climate change

### **Environmental Impact of Forest Depletion on Society**

- Deforestation has a negative impact on society, including loss of biodiversity and increased greenhouse gas emissions
- Efforts are being made to reduce deforestation, including reforestation and sustainable forestry practices

### **Conclusion**

In this unit, we discussed a variety of natural resources that are essential to our survival and the sustainability of our planet. We covered water resources, mineral resources

## **Unit 3 - Pollution and their Effects**

Pollution is the introduction of harmful substances or products into the environment, which can cause various negative effects on human health and the environment. In this unit, we will discuss the different types of pollution and their effects on public health.

### **Public Health Aspects of Environmental Pollution**

Environmental pollution can have significant impacts on public health. Exposure to pollutants can cause a range of health problems, including respiratory and cardiovascular diseases, cancer, and neurological disorders. The following are some of the public health aspects of environmental pollution:

- Increased risk of respiratory diseases such as asthma, bronchitis, and emphysema due to exposure to air pollution.
- Increased risk of cardiovascular diseases such as heart attacks and stroke due to exposure to air pollution.
- Increased risk of cancer due to exposure to environmental pollutants such as asbestos, benzene, and other carcinogens.
- Impacts on children's health, including developmental delays, cognitive impairment, and behavioral problems due to exposure to lead, mercury, and other toxic substances.

- Increased risk of neurological disorders such as Parkinson's disease and Alzheimer's disease due to exposure to environmental pollutants.
- Impacts on reproductive health, including infertility and birth defects due to exposure to environmental pollutants.

### **Water Pollution**

Water pollution is the contamination of water bodies such as lakes, rivers, oceans, and groundwater by pollutants. The following are some of the effects of water pollution:

- Contaminated water can cause waterborne diseases such as cholera, dysentery, and typhoid fever.
- Polluted water can harm aquatic life and disrupt the ecosystem.
- Contaminated groundwater can affect drinking water sources and pose a risk to public health.

### **Air Pollution**

Air pollution is the presence of harmful substances in the air, which can have negative effects on human health and the environment. The following are some of the effects of air pollution:

- Exposure to air pollution can cause respiratory and cardiovascular diseases.
- Air pollution can harm the environment by causing acid rain, ozone depletion, and global warming.
- Exposure to air pollution can have adverse effects on children's health and development.

### **Soil Pollution**

Soil pollution is the contamination of soil by pollutants such as heavy metals, pesticides, and other toxic substances. The following are some of the effects of soil pollution:

- Contaminated soil can harm plants and crops, affecting agricultural productivity.
- Soil pollution can cause health problems for humans and animals that come into contact with contaminated soil.
- Soil pollution can harm the environment by contaminating groundwater and other water sources.

### **Noise Pollution**

Noise pollution is the presence of excessive or unwanted noise, which can have negative effects on human health and well-being. The following are some of the effects of noise pollution:



- Exposure to excessive noise can cause hearing loss and other auditory problems.
- Noise pollution can cause sleep disturbances and affect mental health.
- Exposure to noise pollution can have adverse effects on children's development and academic performance.

### **Solid Waste Management**

Solid waste management is the process of collecting, transporting, and disposing of waste materials. The following are some of the issues related to solid waste management:

- Improper disposal of solid waste can lead to environmental pollution and health hazards.
- Landfills, which are commonly used for solid waste disposal, can generate methane gas, a potent greenhouse gas that contributes to global warming.
- Effective solid waste management strategies, such as recycling and composting, can help reduce the amount of waste that ends up in landfills.

In conclusion, pollution is a significant public health issue that can have adverse effects on human health and the environment. Understanding the different types of pollution and their effects is essential for developing effective strategies to address this issue.

## **Unit 4 - Current Environmental Issues of Importance**

In this unit, we will be discussing some of the most pressing environmental issues that our planet is currently facing. It is important to understand these issues in order to be able to take action and make a positive impact on the environment.

### **Global Warming**

- Global warming is the gradual increase in the Earth's surface temperature.
- It is primarily caused by the release of greenhouse gases, such as carbon dioxide and methane, into the atmosphere.
- This increase in temperature can have a range of negative impacts on the environment, including melting of ice caps, rising sea levels, and more frequent and severe weather events.

### **Greenhouse Effects**

- The greenhouse effect is the process by which certain gases in the Earth's atmosphere trap heat from the sun, leading to a warming of the planet.
- Human activities, such as burning fossil fuels and deforestation, have led to an increase in the levels of these greenhouse gases, exacerbating the greenhouse effect and contributing to global warming.

### **Climate Change**

- Climate change refers to the long-term changes in the Earth's climate, including changes in temperature, precipitation, and weather patterns.
- These changes are largely driven by the increase in greenhouse gases caused by human activities.
- Climate change can have a range of negative impacts on the environment and human societies, including more frequent and severe weather events, food and water shortages, and displacement of people from their homes.

### **Acid Rain**

- Acid rain is a type of rain that has a higher acidity than normal, typically caused by the release of sulfur dioxide and nitrogen oxides into the atmosphere.
- Acid rain can have negative impacts on plant and animal life, as well as on infrastructure such as buildings and bridges.

### **Ozone Layer Formation and Depletion**

- The ozone layer is a layer of gas in the Earth's atmosphere that helps to protect us from harmful ultraviolet radiation from the sun.
- Human activities, such as the release of chlorofluorocarbons (CFCs) into the atmosphere, have led to a depletion of the ozone layer, increasing the risk of skin cancer and other negative health effects.

### **Population Growth**

- The world's population is growing at an unprecedented rate, putting increasing pressure on the Earth's resources and leading to a range of environmental issues.
- As the global population continues to grow, it becomes increasingly important to find sustainable ways to meet our needs without compromising the health of the planet.

### **Automobile Pollution**

- Automobiles are a major source of air pollution, releasing a range of harmful pollutants into the atmosphere, including carbon monoxide, nitrogen oxides, and particulate matter.
- This pollution can have negative impacts on human health, including respiratory problems and increased risk of heart disease.

### **Burning of Paddy Straw**

- Burning of paddy straw is a common practice in agricultural areas, but it can have negative impacts on the environment and human health.

- The smoke from burning straw can contribute to air pollution, and the burning of straw can also contribute to soil degradation and loss of nutrients.

In conclusion, these environmental issues are complex and multifaceted, but it is important for us to understand them in order to take action and make a positive impact on our planet. By working together and taking steps to reduce our impact on the environment, we can help to ensure a sustainable future for ourselves and future generations.

## **Unit 5 - Environmental Protection**

This unit covers various aspects of environmental protection, including the Environmental Protection Act 1986, initiatives by Non-Governmental Organizations (NGOs), and the impact of human population on the environment.

### **Environmental Protection Act 1986**

The Environmental Protection Act 1986 is an important law in India that aims to protect and improve the quality of the environment. Some key features of the act include:

- The act provides for the prevention, control, and abatement of environmental pollution.
- It establishes various authorities responsible for implementing the provisions of the act, including the Central Pollution Control Board and State Pollution Control Boards.
- The act also provides for penalties and punishments for those who violate its provisions.

### **Initiatives by Non-Governmental Organizations (NGOs)**

NGOs play a crucial role in promoting environmental protection by raising awareness, advocating for policy changes, and implementing projects at the grassroots level. Some notable initiatives by NGOs include:

- The Chipko movement, which started in the 1970s and aimed to protect forests by hugging trees to prevent them from being cut down.
- The Narmada Bachao Andolan, which opposed the construction of large dams on the Narmada river and advocated for the rights of local communities.
- The Greenpeace campaign against the use of plastic bags, which has helped to raise awareness about the impact of plastic pollution on the environment.

## **Human Population and the Environment**

Human population growth has a significant impact on the environment, including increasing demand for resources and contributing to pollution and climate change. Some important aspects of this issue include:

- Population growth puts pressure on natural resources such as water, forests, and land.
- Environmental education can help to raise awareness about the impact of human activities on the environment and promote sustainable practices.
- Women's education is also important, as it has been shown to be associated with lower fertility rates and better reproductive health outcomes.

In conclusion, environmental protection is a crucial issue that requires a multifaceted approach. Laws such as the Environmental Protection Act 1986 and initiatives by NGOs play an important role in promoting environmental protection, while addressing the impact of human population growth on the environment requires a combination of education and policy changes.